

Module 03: Perception in Robotics

Learning Objectives

1. Define **Perception** within the context of Robotics.
2. Analyze how Perception interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Perception.

Introduction

This module explores **Perception**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Perception is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

Key Concepts

1. Perception as a Markov Blanket Boundary

How does Perception define the boundary between the agent and the environment?

2. Generative Models of Perception

What parameters involved in Perception must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Perception drive the perception-action loop?

Applications

In Robotics, we see Perception manifest in: * **Specific Example 1:** [Add domain-specific example here] * **Specific Example 2:** [Add domain-specific example here]

Conclusion

Understanding Perception allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.